

Optimal Semi-Fragile Watermarking Based on Maximum Entropy Random Walk and Swin Transformer for Tamper Localization

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Abstract

Verifying digital audio authenticity requires not only detecting modifications but also precisely localizing tampered regions. This paper proposes an optimal semi-fragile audio watermarking system utilizing Maximum Entropy Random Walk (MERW) graph modeling and Swin Transformer architecture. The watermark key is generated dynamically from low-frequency DWT-SVD coefficients, while Swin Transformer extracts high-level acoustic feature representations to classify untouched vs manipulated segments.

Key Methodology & Architecture

Discrete Wavelet Transform (DWT), Singular Value Decomposition (SVD), MERW graph optimization, Swin Transformer network for local spatial-temporal feature attention.

Datasets & Benchmark Environments

TIMIT Acoustic-Phonetic Continuous Speech Corpus, Free Music Archive (FMA).

Performance Metrics & Graph Axis Parameters

Reported Results: Tamper Localization Accuracy = 99.4%, Precision = 99.1%, Recall = 99.6%, Normalized Cross-Correlation (NC) = 0.998.

Graph Parameter Mapping: X-axis: Tampering Ratio (% of audio altered, 0% to 50%); Y-axis: Tamper Localization Precision & Recall (0.0 to 1.0).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Extremely fine millisecond-level tamper localization, high fidelity preservation.	Requires high computational GPU memory for Swin Transformer inference.

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